Course Project Overview

Over this semester you will design and carry out an applied deep learning project with a research-oriented structure. This is not a "download a dataset, train a model, report accuracy" assignment. You will choose a problem worth solving, build a baseline, build something stronger, run controlled experiments, and explain what you found — including what did not work.

Project workflow

Your project should follow approximately this path:

Problem → Dataset → Baseline → Deep Learning Model → Experiments → Evaluation → Interpretation → Application

Technical requirements

Requirement	Detail
Framework	All neural networks must be implemented and trained using **TensorFlow / Keras** (`tf.keras`)
Environment	**Kaggle Notebooks with GPU acceleration**
GPU verification	Show `tf.config.list_physical_devices('GPU')` returning a GPU in your notebook
Supporting libraries	NumPy, Pandas, Matplotlib, and scikit-learn are permitted for preprocessing, visualization, evaluation, and supporting analysis

Your problem should be substantial enough that GPU acceleration provides a meaningful benefit.

Dataset expectations

You choose your own dataset. It must be substantial enough to support a meaningful deep learning project. Three categories are preferred:

Option 1 — Computer Vision. Medical images, satellite or aerial imagery, agricultural or environmental images, disaster imagery, industrial inspection, wildlife, large image classification sets, segmentation, or object detection.

> MNIST and Fashion-MNIST are **not** appropriate for this project.

Option 2 — Large Text or Sequence Data. Student responses, reviews, social-media posts, educational interaction logs, documents, conversations, scientific text, large corpora, or time-series and sequential interaction data. Tasks might include text classification, multi-label classification, sequence prediction, knowledge tracing, emotion classification, or document classification.

Option 3 — Multimodal Data. Image + text, image + metadata, video + text, or educational interactions + learner information. A multimodal project might compare **Image Only → Text Only → Combined Model**.

Research-oriented dataset sources

You are encouraged to use datasets tied to real research problems. Good sources include Kaggle, Hugging Face Datasets, TensorFlow Datasets, GitHub repositories attached to academic papers, university research laboratories, public research datasets, government open-data repositories, and datasets linked from conference or journal papers.

One example is the Purdue CourseMIRROR Reflection Quality dataset:
`https://engineering.purdue.edu/coursemirror/download/reflections-quality-data/`

Where possible, choose a dataset that provides documentation, citation information, an associated research paper, usage or licensing information, and enough data to support deep learning. **If your dataset was introduced through an academic publication, read and cite the original dataset paper.**

Experimentation requirement

Run controlled experiments — change **one important factor at a time** so you can explain why performance changed. Examples: learning rate, optimizer, batch size, architecture depth, dropout, L1/L2 regularization, data augmentation, transfer learning, scratch vs. pretrained, frozen vs. fine-tuned layers, class weighting, sampling strategy, architecture comparison (CNN vs. LSTM, LSTM vs. Transformer), unimodal vs. multimodal, robustness, explainability, or model efficiency.

Research / applied extension

Every final project must include at least one element beyond training and evaluating a standard model:

- **Explainability** — Grad-CAM or another appropriate technique
- **Robustness** — performance under blur, noise, poor lighting, rotation, cropping, compression, or another meaningful perturbation
- **Transfer learning** — Scratch → Pretrained → Fine-tuned
- **Class imbalance** — approaches for minority classes
- **Multimodal learning** — individual modalities vs. combined
- **Model efficiency** — performance vs. parameter count, training time, inference time
- **Domain generalization** — train on one domain, test on another
- **Architecture comparison** — multiple appropriate architectures

This does not need to be a novel research contribution. The goal is **systematic experimentation and research-oriented thinking**.

Evaluation and error analysis

Use metrics appropriate to your problem — Accuracy, Precision, Recall, F1, Macro F1, ROC-AUC, Confusion Matrix, IoU, Dice, MAE, RMSE. Do not rely on accuracy alone when other metrics carry important information.

Your final project must include meaningful error analysis of roughly **5–10 interesting failure cases**, discussing what the model got wrong, whether errors form patterns, which classes are hardest, and whether the dataset, preprocessing, or architecture contributed.

Applied demonstration

Show that your trained model works on new data — upload an image and return a prediction, enter text and receive a prediction, generate a Grad-CAM visualization, process new examples, or build a small Gradio demonstration. A production-ready application is not required.

Formatting Style: APA 7th Edition

All formal written Capstone submissions follow **APA 7th Edition**.

Graduate students

Graduate students have one additional requirement: a short **Related Work** section, approximately half a page with a minimum of three scholarly papers, included in the Final Capstone Report. This accounts for **20% of the graduate student's Capstone Project grade**. There is no separate deadline — it is submitted as part of the Final Report. All other requirements are identical for graduate and undergraduate students.

Timeline

Assignment / Milestone	Release Date	Due Date
Capstone Part 1	September 14, 2026	October 22, 2026
Capstone Part 2	—	November 12, 2026
Capstone Part 3 — Final Presentation	—	December 1 & 3, 2026
Final Capstone Report	—	December 10, 2026

What a strong project looks like

A strong project has a clear question someone would actually want answered, a dataset substantial enough to support it, a baseline you can beat and a stronger model that beats it, experiments where one factor changed at a time so the results mean something, evaluation metrics chosen for the problem rather than by habit, honest error analysis, and a clear account of limitations. A project that reports a modest result and explains it well will score better than one that reports a high number it cannot account for.